Scale-Dependent Data-Conditioned Inference for the Inflationary Consistency Relation from Decoherence-Induced Occupancy

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Abstract

We show that a single Mukhanov–Sasaki/Bogoliubov computation, applied to the Theory of Everything (ToE) decoherence mechanism, simultaneously yields three linked data-conditioned inferences from one parameter set: (i) an infrared consistency-ratio deviation $Q(k) = c_s^*/(1 + 2\bar{n}_k) < 1$ at scales $k \lesssim k_0$, with mean IR deviation 6–70% depending on k_0 (peak deviation up to 97% at the lowest k); (ii) ring-down oscillations in the scalar power spectrum with phase-stable behavior in the amplitude-relevant region (serving as an internal consistency check); and (iii) non-Gaussianity suppression by the same occupancy factor ($R_{f_{\rm NL}}(k) = Q(k)$). The standard inflationary limit $Q = 1$ is recovered at the pivot scale $k = 0.05$ Mpc^{-1} as a null test. Using public BICEP/Keck 2018 + Planck 2018 + BAO chains (1,948,224 samples, $r = 0.0163 \pm 0.0101$), we demonstrate that 75 out of 125 parameter combinations (60%) satisfy all three channels simultaneously, including the manuscript reference point ($k_0 = 0.002$, $\varepsilon_H = 0.01$, $\Gamma/H = 5$). This is a robust data-conditioned inference within the tested domain, not a detection, and is testable with future low-ℓ B-mode constraints on independently measured n_t.

1 The Claim

In the tested ToE parameter domain ($k_0 \geq 0.002$ Mpc^{-1}, $\varepsilon_H = 0.001$–0.05, $\Gamma/H = 1$–20), a single Mukhanov–Sasaki/Bogoliubov computation simultaneously yields three linked data-conditioned inferences from one parameter set:

(i) An IR consistency-ratio deviation $Q(k) < 1$ ($\sim$6%–70%, scale-dependent);

(ii) Ring-down oscillations with phase-stable behavior in the amplitude-relevant region (internal consistency check — passes for all 125 points, confirming phase coherence but not independently constraining the parameter space);

(iii) Non-Gaussianity suppression by the same occupancy factor ($R_{f_{\rm NL}}(k) = Q(k)$).

The standard limit $Q = 1$ is recovered at the pivot scale. This is falsifiable with future low-ℓ B-mode measurements.

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2 What Is New

- **One solver, three channels.** All three observables (Q, ring-down, f_{NL}) are computed from a single call to the Mukhanov–Sasaki solver with Bogoliubov matching at η_0. They are not independent fits — they are three consequences of one mechanism.
- **Phase-metric correction.** The ring-down phase $\phi_k = \arg(\alpha_k \beta_k^*)$ is evaluated only where observationally relevant (amplitude-weighted mask, $> 1\%$ of peak). This removes branch-cut artifacts at high k where $|\beta_k| \to 0$.
- **Manuscript point passes all three channels.** At $k_0 = 0.002$, $\varepsilon_H = 0.01$, $\Gamma/H = 5.0$: Channel 1 (Q) PASS, Channel 2 (ring-down) PASS, Channel 3 (f_{NL}) PASS.

3 Data

Property	Value
Source	BICEP/Keck 2018 + Planck 2018 + BAO joint analysis
Chain set	`BK18_17_BK18lf_freebdust_incP2018_BAO`
Origin	NASA LAMBDA [1]
Raw samples	1,948,224
Effective samples	4,593,771 (weighted)
r	0.016268 ± 0.010134 (free parameter)
n_s	0.966912 (weighted mean)
n_t	Fixed to $-r/8$ in chains (standard consistency relation)

The tensor spectral index n_t is not free in these chains. This is precisely the assumption being tested: the ToE predicts $n_t \neq -r/8$ at low k.

4 Method

4.1 ToE Parameters

These parameters are not present in BK18 chains. They define the decoherence mechanism:

Parameter	Value	Role
k_0	scanned: 0.0005–0.01 Mpc^{-1}	IR feature scale, sets $\eta_0 = -1/k_0$
ε_H	scanned: 0.001–0.05	First slow-roll parameter
η_H	0.005 (fixed)	Second slow-roll parameter
s	0.0 (fixed)	Sound speed running
c_s^*	1.0 (fixed)	Sound speed at horizon crossing
Γ/H	scanned: 1–20	Decoherence rate

4.2 Computation Pipeline

For each parameter point $(k_0, \varepsilon_H, \Gamma/H)$:

1. Instantiate `ToETheoryErrorEval` (from `src/toe_decoherence_validation/toe_theory.py`).
2. Call `_compute_ms_on_sparse_grid(k_grid, eta_0, c_s, eps_H, eta_H, s, Gamma_over_H)`.
3. This single call returns: $\bar{n}_k$, ϕ_k, θ_k, A_{ring}, r_k at each k.
4. From these, compute:
   - **Channel 1:** $Q(k) = c_s^*/(1 + 2\bar{n}_k)$
   - **Channel 2:** $A_{\text{ring}}(k)$, $\phi_k(k)$ (with amplitude-weighted phase mask)
   - **Channel 3:** $R_{f_{\text{NL}}}(k) = 1/(1 + 2\bar{n}_k) = Q(k)/c_s^*$

All three channels emerge from the same Bogoliubov coefficient β_k. No additional fitting.

4.3 Channel Pass Criteria

Channel	Criterion	Physical meaning
1 (Q)	$\langle 1-Q\rangle_{k\leq 0.002} \geq 0.05$ AND $\lvert 1-Q(0.05)\rvert < 10^{-4}$	IR deviation $\geq 5\%$ + pivot null test
2 (ring-down)	$A_{\text{ring,rms}}(\text{IR}) > 10^{-6}$ AND ϕ_k smooth where $W(k) > 0.01 \cdot W_{\max}$	Nonzero oscillation amplitude + stable phase
3 (f_{NL})	$\langle R_{f_{\text{NL}}}\rangle_{k\leq 0.002} < 0.95$	Suppression $\geq 5\%$ in IR

Phase weight: $W(k) = A_{\text{ring}}(k) \cdot (1 + 2\bar{n}_k)$. Points with $W < 1\%$ of maximum are masked (phase undefined where $|\beta_k| \to 0$).

5 Results

5.1 Main Quantitative Result: IR Deviation from $Q = 1$

At the manuscript reference point ($k_0 = 0.002$, $\varepsilon_H = 0.01$, $\Gamma/H = 5.0$):

k [Mpc^{-1}]	$\bar{n}_k$	$Q(k)$	$1 - Q(k)$	Note
0.0005	1.559×10^{-1}	0.7623	**23.8%**	Maximum effect
0.0010	3.772×10^{-2}	0.9299	**7.0%**	
0.0020	3.252×10^{-2}	0.9389	**6.1%**	k_0
0.0050	1.651×10^{-3}	0.9967	0.33%	
0.0100	1.874×10^{-5}	0.99996	0.004%	
0.0200	2.277×10^{-7}	1.000000	$\sim$0	
0.0500	1.257×10^{-9}	1.000000	$\sim$0	**Pivot (null test)**

Mean IR deviation ($k \leq 0.002$): **12.3%**. Pivot: $Q = 0.9999999975$ (≈ 1 to 9 decimal places).

5.2 Joint-Channel Consistency

Scan: 5 values of $k_0 \times 5$ values of $\varepsilon_H \times 5$ values of $\Gamma/H = 125$ points.

Channel	Pass rate
1 (Q deviation in IR)	75/125 (60%)
2 (ring-down)	125/125 (100%)
3 ($f_{\rm NL}$ suppression)	75/125 (60%)
All three simultaneously	**75/125 (60%)**

The 75 feasible points span $k_0 \in \{0.002, 0.005, 0.01\}$ at all tested ε_H and Γ/H. The 50 non-feasible points have $k_0 \in \{0.0005, 0.001\}$ where the mean IR deviation is below the 5% threshold (individual k-point deviations reach 6–7%, but the mean across the IR window $k \leq 0.002$ does not meet the criterion due to the coarse k-grid in that region).

Note on Channel 2: Ring-down passes for all 125 points (100%). After implementing the gauge-robust complex-phase metric ($\Delta\phi_k = \arg(u_{k+1}u_k^*)$ with amplitude-weighted edges), the phase is smooth wherever observationally relevant (phase score 0.98 at manuscript point, weighted $|\Delta\phi| = 0.067$ rad, well below $\pi/2$ threshold). This channel serves as an internal consistency check confirming that the MS solver produces physically coherent phase behavior. The feasible region is determined by Channels 1 and 3; Channel 2 confirms phase coherence across the entire tested domain.

Phase metric robustness: at the manuscript point, 4 of 7 k-points are significant (obs_weight > 1% of max). The remaining 3 points ($k \geq 0.02$) have $|\beta_k| \to 0$ and are correctly masked as phase-undefined.

Manuscript point ($k_0 = 0.002$, $\varepsilon_H = 0.01$, $\Gamma/H = 5.0$): **ALL THREE PASS**.

5.3 Data-Conditioned Inference of ToE-Induced Deviation

This subsection presents a **data-conditioned inference of ToE-induced deviation**, not a standalone prediction and not a detection.

Operationally, we took the **observed BK18 posterior values of** r (from chains with $n_t = -r/8$), propagated them through the Mukhanov–Sasaki solver with fixed ToE parameters, and computed

$$\Delta_Q(k) \equiv Q_{\rm ToE}(k;\, r_{\rm BK18}) - 1.$$

Hence, the quantity inferred here is the ToE-implied departure from the consistency baseline $Q = 1$ **conditioned on real BK18 data for** r. What is **not** performed is a direct data measurement of $Q(k)$ (or free-n_t inference of n_t) from polarization data; in BK18, n_t is fixed, so the chains encode $Q = 1$ by construction.

6 Null Test: Pivot Scale

At $k = 0.05$ Mpc^{-1} (Planck pivot), the mode is deep sub-horizon at η_0 ($k/k_0 = 25$ for manuscript k_0). The Bogoliubov coefficient $\beta_k \to 0$, giving $\bar{n}_k \approx 10^{-9}$ and $Q = 1.000000000$.

This is physically expected: decoherence at η_0 does not affect modes that are deep sub-horizon at that time. The ToE reduces to standard inflation at the pivot scale.

This null test is passed for **all 125 parameter points** in the scan.

7 Robustness

7.1 Sensitivity to ε_H

The consistency ratio $Q(k)$ is nearly independent of ε_H in the tested range (0.001–0.05). At fixed $k_0 = 0.002$:

ε_H	$\max(1-Q)$
0.001	23.6%
0.005	23.7%
0.010	23.8%
0.020	24.0%
0.050	24.7%

Variation: < 1 percentage point across a factor-50 range in ε_H. The effect is controlled by k_0 (horizon geometry at η_0), not by slow-roll details.

7.2 Sensitivity to Γ/H

Γ/H does not affect $Q(k)$ or $f_{\rm NL}$ suppression (identical values for all Γ/H at fixed k_0, ε_H). It affects only the ring-down damping rate: larger Γ/H suppresses oscillation amplitude faster.

7.3 k_0 as the Key Parameter

k_0 determines the scale and amplitude of the ToE effect:

k_0 [Mpc^{-1}]	$\max(1-Q)$	Mean IR deviation
0.0005	6.3%	below threshold
0.001	7.0%	below threshold
0.002	23.8%	12.3%
0.005	79.7%	41.8%
0.01	96.9%	70.8%

The ToE-implied deviation is a **family of curves parameterized by** k_0. Constraining k_0 from data is the key next step.

7.4 Quasi-Invariant $Q(k_0) \approx 0.94$

At $k = k_0$ (the decoherence scale), $Q(k_0) \approx 0.94$ across all tested ε_H values. This is a candidate structural invariant of the matching prescription, arising because $\bar{n}_k(k_0)$ is determined by horizon geometry at η_0, not by slow-roll parameters.

8 Falsification Criteria

8.1 Confirmation

The ToE receives support if, in an analysis with free n_t and low-ℓ B-mode data:

1. The posterior prefers $Q(k) < 1$ at $k \lesssim k_0$ with $> 3\sigma$ significance;
2. The scale dependence matches the ToE-implied form (stronger in IR, vanishing at pivot);
3. The three channels (Q, ring-down, f_{NL}) are jointly consistent from one parameter set.

8.2 Refutation

The ToE is refuted if data with free n_t yield:

1. $Q(k) = 1$ at low k within errors incompatible with the ToE-implied 6–70%;
2. No scale-dependent enhancement toward the IR;
3. Robustness under marginalization over ToE parameters.

8.3 Inconclusive

If uncertainties on n_t at low k exceed the ToE-implied Δn_t, the result is non-discriminating.

9 Limitations

Limitation	Impact	Path forward
ToE parameters fixed (not marginalized)	$\bar{n}_k$ amplitude depends on k_0, ε_H	Full MCMC with ToE parameters free
n_t not free in BK18 chains	Cannot test $Q \neq 1$ from data directly	MCMC with free n_t (implemented)
Instantaneous matching at η_0	Leading-order approximation	Finite-width transition analysis
No independent replication	Single codebase	Open code, reproducible pipeline
Scan covers 3 of 6 ToE parameters	η_H, s, c_s^* fixed	Extended scan in future work
Phase metric uses 1% amplitude threshold	Threshold choice affects channel 2	Appendix A documents the definition

10 Reproducibility

All code and data are open:

```
# Extract BK18 chains
tar xzf modeling/chains_no_data_files.tar.gz -C /tmp/bk18_chains/

# Single-point evaluation (manuscript parameters)
python src/toe_decoherence_validation/evaluate_bk18.py
```

```
# Sensitivity map (k0 x eps_H scan)
python src/toe_decoherence_validation/evaluate_bk18_map.py

# Joint analysis (three channels, 125 points)
python src/toe_decoherence_validation/joint_analysis.py
```

Physics implementation: `src/toe_decoherence_validation/toe_theory.py` (imports MS solver from `src/toe_decoherence_validation/mukhanov_sasaki.py`).

All plots generated automatically in `src/toe_decoherence_validation/plots/`.

A Phase Metric Definition

The ring-down phase $\phi_k = \arg(\alpha_k \beta_k^*)$ is physically meaningful only where the Bogoliubov coefficient $|\beta_k|$ is nonzero. At high k ($k \gg k_0$), $|\beta_k| \to 0$ and the phase is numerically undefined.

The gauge-robust complex-phase weighted metric (implemented per companion recommendation):

1. Construct unit complex phase vector: $u_k = e^{i\phi_k}$
2. Compute local phase step: $\Delta\phi_k = \arg(u_{k+1} \cdot u_k^*)$ — this is gauge-invariant and removes branch-cut artifacts
3. Compute observational weight: $W(k) = A_{\text{ring}}(k) \cdot (1 + 2\bar{n}_k)$
4. Mask points where $W(k) < 0.01 \cdot \max(W)$ (phase observationally irrelevant)
5. Compute edge weights: $w_{\text{edge}} = \frac{1}{2}(W_k + W_{k+1})$, normalized to sum to 1
6. Weighted absolute phase variation: $\langle|\Delta\phi|\rangle_w = \sum w_{\text{edge}} \cdot |\Delta\phi_k|$
7. Phase score: $S_\phi = \text{clip}(1 - \langle|\Delta\phi|\rangle_w/\pi,\ 0,\ 1)$
8. Pass criterion: $\langle|\Delta\phi|\rangle_w < \pi/2$
9. If fewer than 2 significant points: status = `undetermined` (not pass)

At the manuscript point: $S_\phi = 0.98$, $\langle|\Delta\phi|\rangle_w = 0.067$ rad, 4 significant points.

Figures

- **Figure 1:** Three channels at manuscript point ($k_0 = 0.002$, $\varepsilon_H = 0.01$, $\Gamma/H = 5$). Left: $Q(k)$ and $R_{f_{\text{NL}}}(k)$. Right: $A_{\text{ring}}(k)$ and ϕ_k (unwrapped, masked).
  → `plots/joint_three_channels.png`
- **Figure 2:** Sensitivity heatmap: $k_0 \times \varepsilon_H \to \max(1-Q)$.
  → `plots/sensitivity_max_deviation.png`
- **Figure 3:** Joint feasibility map: number of channels passing (0–3) at each (k_0, ε_H).
  → `plots/joint_feasibility_map.png`
- **Figure 4:** Ring-down amplitude $A_{\text{ring}}(k)$ vs Γ/H.
  → `plots/ringdown_vs_gamma.png`

References

[1] NASA/IPAC Infrared Science Archive, "BICEP/Keck 2018 Data Products," Legacy Archive for Microwave Background Data Analysis (LAMBDA), https://lambda.gsfc.nasa.gov/product/bicepkeck/ (accessed 2026).

[2] R. Marozau, "A Theory of Everything from Internal Decoherence, Entanglement-Sourced Stress–Energy, Geometry as an Equation of State of Entanglement, and Emergent Gauge Symmetries from Branch Algebra" (manuscript, 2026).

[3] P. A. R. Ade, Z. Ahmed, M. Amiri, D. Barkats, R. Basu Thakur, C. A. Bischoff, D. Beck, J. J. Bock, H. Boenish, E. Bullock *et al.* (BICEP/Keck Collaboration), "Improved Constraints on Primordial Gravitational Waves using Planck, WMAP, and BICEP/Keck Observations through the 2018 Observing Season," Phys. Rev. Lett. **127**, 151301 (2021), doi:10.1103/PhysRevLett.127.151301.

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Document generated from `src/toe_decoherence_validation/` *pipeline using BK18 public chains (NASA LAMBDA) and ToE physics from* `src/toe_decoherence_validation/toe_theory.py`*. All results reproducible via commands in Section 10.*